

	Validation sets of escape mutations per domains
NTD	(1) L18F, (2) T19R, (3) A67V, (4) D80A, (5) T95I, (6) G142D, (7) L212I, (8) D215G
RBD	(9) G339D, (10) S371L, (11) S373P, (12) S375F, (13) K417E, (14) K417N, (15) N440K, (16) N440Q, (17) V445A, (18) G446S, (19) N450D, (20) L452R, (21) Y453F, (22) L455F, (23) S477N, (24) T478K, (25) E484A, (26) E484K, (27) G485D, (28) F486V, (29) F490L, (30) F490S, (31) Q493K, (32) Q493R, (33) G496S, (34) Q498R, (35) N501Y, (36) Y505H
FP	(52) D796Y
HR1	(54) D950N, (55) Q954H, (56) N969K, (57) L981F, (58) S982A
HR2	
TM	
Others (Not in domains)	(37) T547K, (38) A670D, (39) D614G, (40) H655Y, (41) N679K, (42) P681H, (43) P681R, (44) R682Q, (45) R685S, (46) V687G, (47) A701V, (48) T716I, (49) N764K, (50) G769E, (51) Q779K, (59) D1118H, (60) V1128A